

About Weights in Deep Learning

In deep learning, **Weights** are numbers that tell the model how important each input is.

How It Works

A model does this:

Output=(input×weight)+b

- b → bias

Why Do Weights Matter?

- They help the model **learn patterns**
- They decide **which input matters more**
- Without weights → model cannot learn anything

How Weights Change

At first:

- Weights are random

During training:

- Model checks error
- Updates weights
- Improves prediction

Types of Weights in Deep Learning

1. Input Layer Weights
 - Connect input → hidden layer
2. Hidden Layer Weights
 - Learn complex patterns
3. Output Layer Weights
 - Final decision making

Xception algorithm in detail

Xception (Extreme Inception) is a deep learning model used for image classification. It improves traditional CNNs by using **depthwise separable convolutions**, which split the learning process into two simple steps:

1. Learning spatial features
2. Learning relationships between channels

Structure

Xception has 3 parts:

1. **Entry Flow** → Learns basic features
2. **Middle Flow** → Learns complex patterns (repeated blocks)
3. **Exit Flow** → Gives final prediction

How It Works

Instead of doing everything at once like normal CNN:

- Step 1: Detect patterns in each channel (edges, shapes)
- Step 2: Combine those patterns

This makes the model:

- Faster
- More efficient
- Better at learning

Key Features

- Uses **depthwise separable convolution**
- Uses skip connections from ResNet
- Needs fewer parameters than normal CNN

Advantages

- High accuracy
- Less computation
- Good for large datasets like ImageNet

Use of model saving in Deep Learning

Model saving in deep learning means storing a trained model (its **weights + architecture**) so you can use it later without training again.

Why is Model Saving Important?

Training a model:

- Takes **a lot of time**
- Uses **CPU/GPU resources**

So instead of training again and again, we **save the model once and reuse it**.

Main Uses of Model Saving

1. Reuse the Model (Inference)

- Use the trained model to make predictions
- Example:
 - Image classification
 - Fraud detection

No need to retrain

2. Save Training Time

- Training deep learning models can take **hours or days**
- Saving avoids repeating the same work

3. Deployment in Real Applications

- Saved models are used in:
 - Mobile apps
 - Websites
 - APIs

Example:

- Face recognition system
- Chatbots

4. Resume Training Later

- You can **stop and continue training**
- Useful if:
 - System shuts down
 - Long training process

5. Model Sharing

- Share model with others
- Used in:
 - Research
 - Kaggle competitions

6. Version Control

- Save different versions of models
- Compare performance

What Gets Saved?

When you save a model:

- Model architecture
- Learned weights
- Training configuration (optional)

Common Formats

- HDF5 (.h5)
- SavedModel (TensorFlow format)
- .pt / .pth (PyTorch)

Code

```
# Save model
```

```
model.save("model.h5")
```

```
# Load model
```

```
from tensorflow.keras.models import load_model
```

```
model = load_model("model.h5")
```